

Lab 4: **Scalable QSVM simulation**

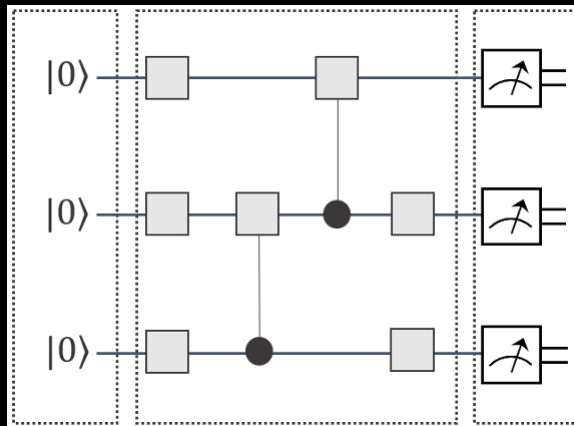
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Quantum Computing Team

National Center for High-performance Computing (NCHC), NIAR

Quantum Circuit Simulator

Quantum Circuit (Gate-Based Quantum Computing)



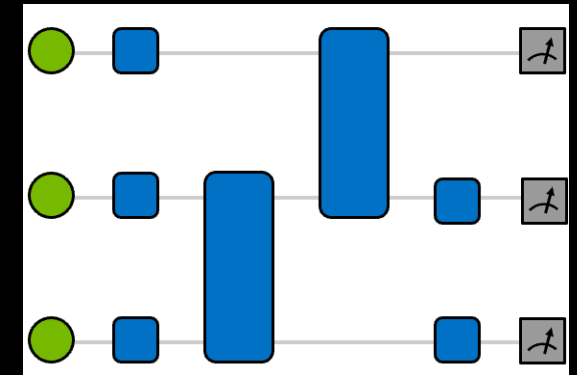
Put **dataset** (Classical or Quantum)
into Quantum Algorithm

Noisy Intermediate-Scale QPU



Limited scale and quality of
qubits and gates
with **insufficient accuracy**

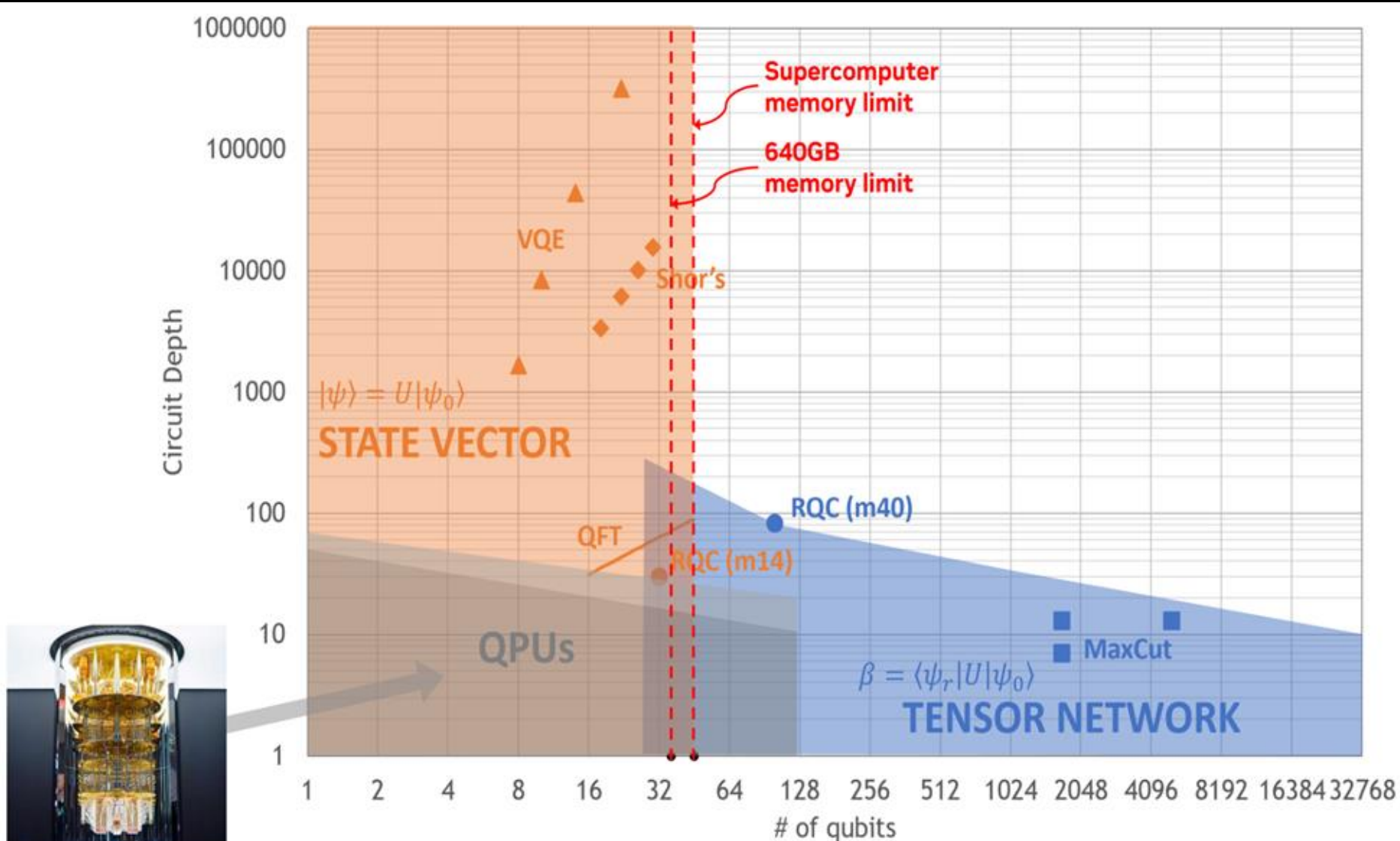
Quantum Circuit Simulation (HPC)



Tensor operations on **CPU/GPUs**
with **accurate results**



Quantum Circuit Simulator



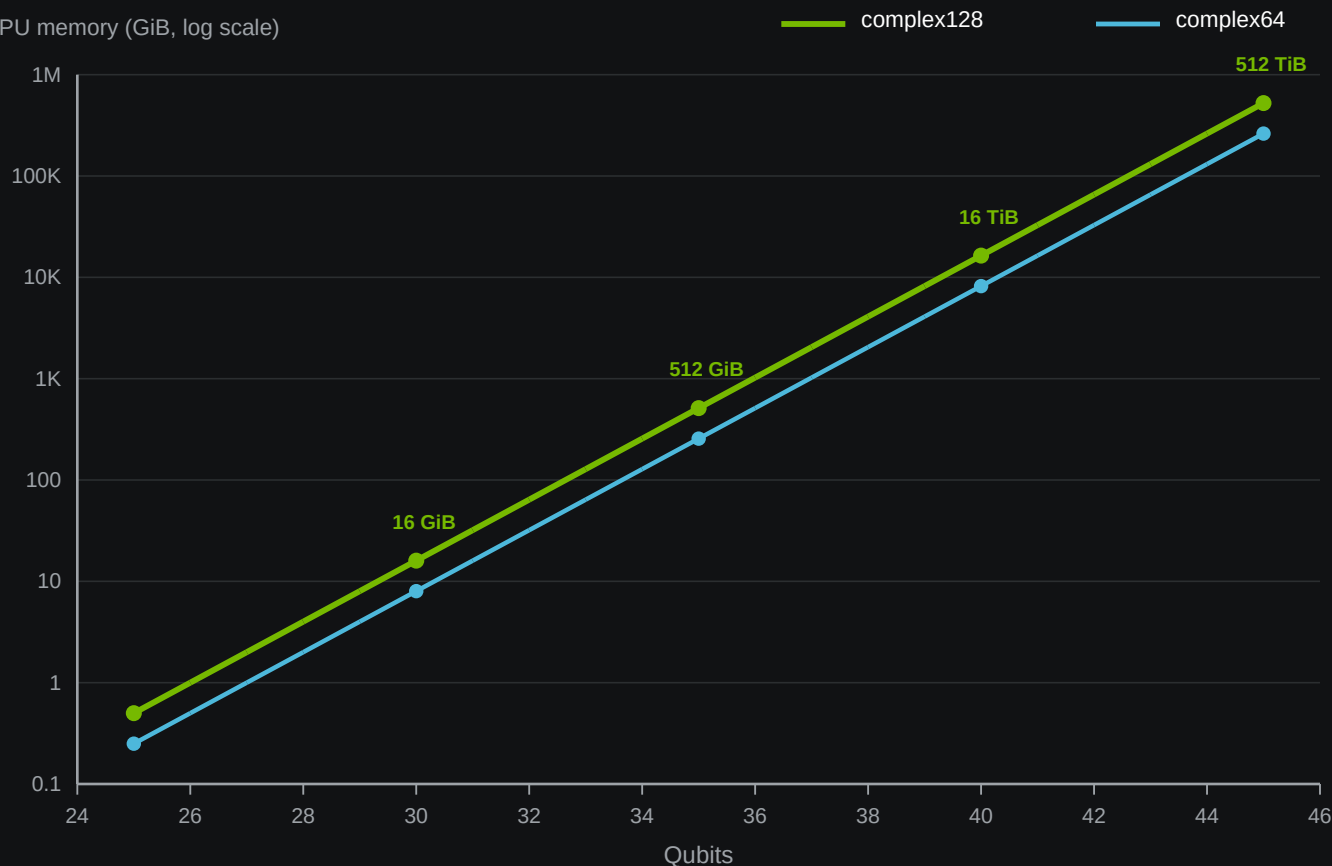
State-Vector Simulation

Exact amplitudes and simple gate updates - with exponential memory growth

Memory scales as 2^n

one complex number / basis state

GPU memory (GiB, log scale)



WHAT THE SIMULATOR STORES

2^n complex amplitudes

FULL-STATE ACCESS

- Any amplitude
- Exact state inspection
- Straightforward gate application

COMPUTE + MEMORY

Memory: $O(2^n)$

Gate work: approximately $O(G \times 2^n)$

THE PRACTICAL LIMIT

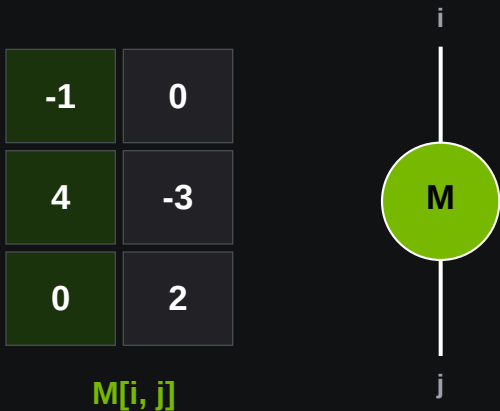
40 qubits = 16 TiB (complex128)

Tensor-Network Simulation

Turn gates into tensors, connect shared indices, then contract only what the result needs

1 REPRESENT

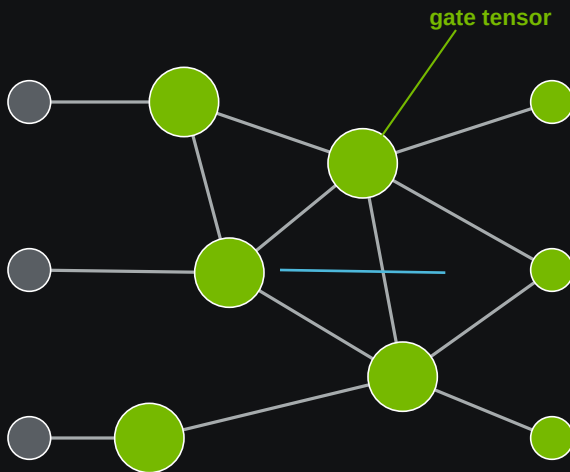
A matrix or gate is a tensor



Node = tensor
Leg = index

2 CONNECT

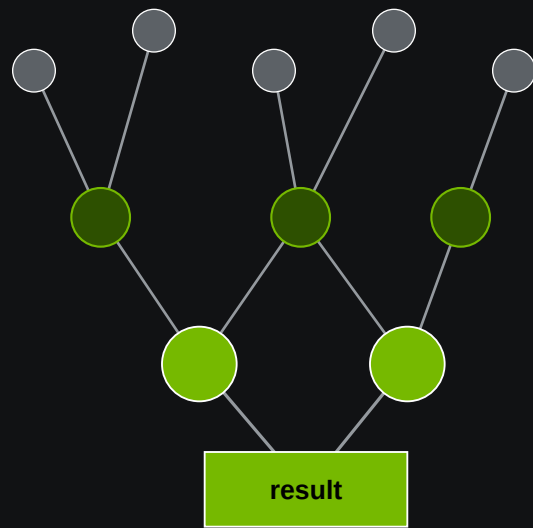
Shared indices become edges



Circuit topology
becomes a graph

3 CONTRACT

Choose an order, sum shared indices

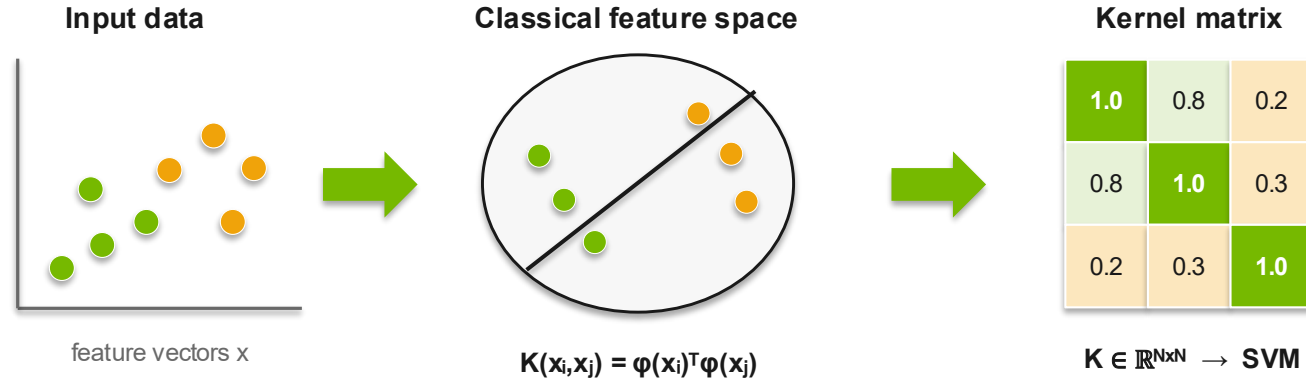


SCALING COMPARISON

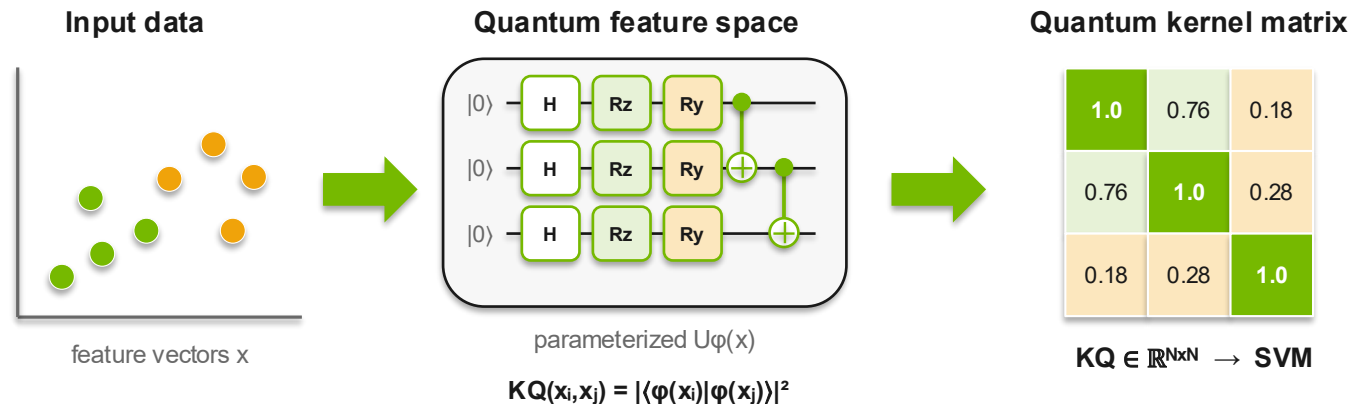
METHOD		TIME	MEMORY	KEY VARIABLE	INTERPRETATION
SV	full state	$O(G \times 2^n)$	$O(2^n)$	n = qubits	Exact access; memory-limited
TN	contract graph	$O(\text{poly}(N) \times 2^w)$	$O(2^w)$	w = contraction width	Advantage when $w \ll n$ Worst case: $w \sim n$

Quantum-Enhanced SVM (QSVM)

Classical SVM (kernel method)



Quantum-enhanced SVM (quantum kernel)

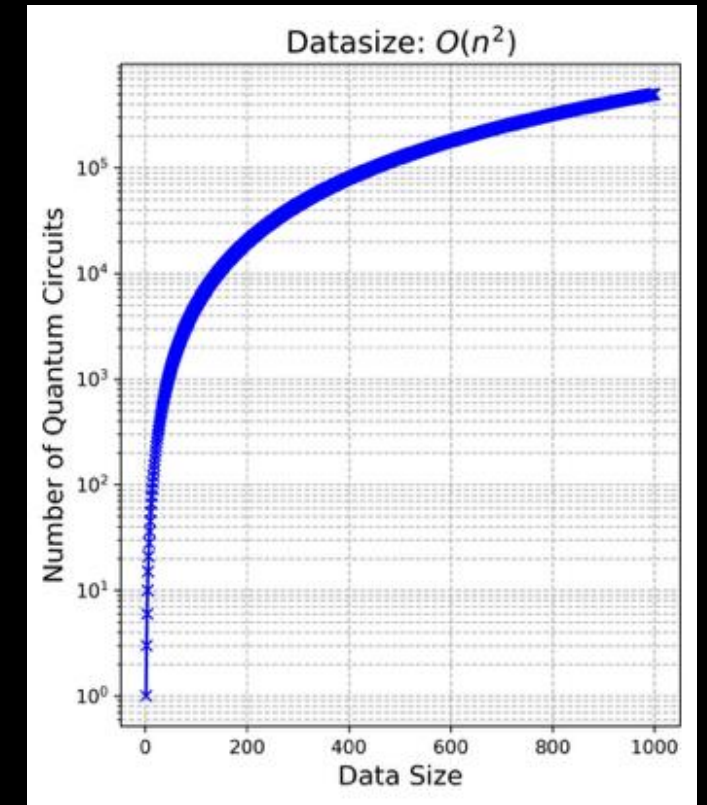
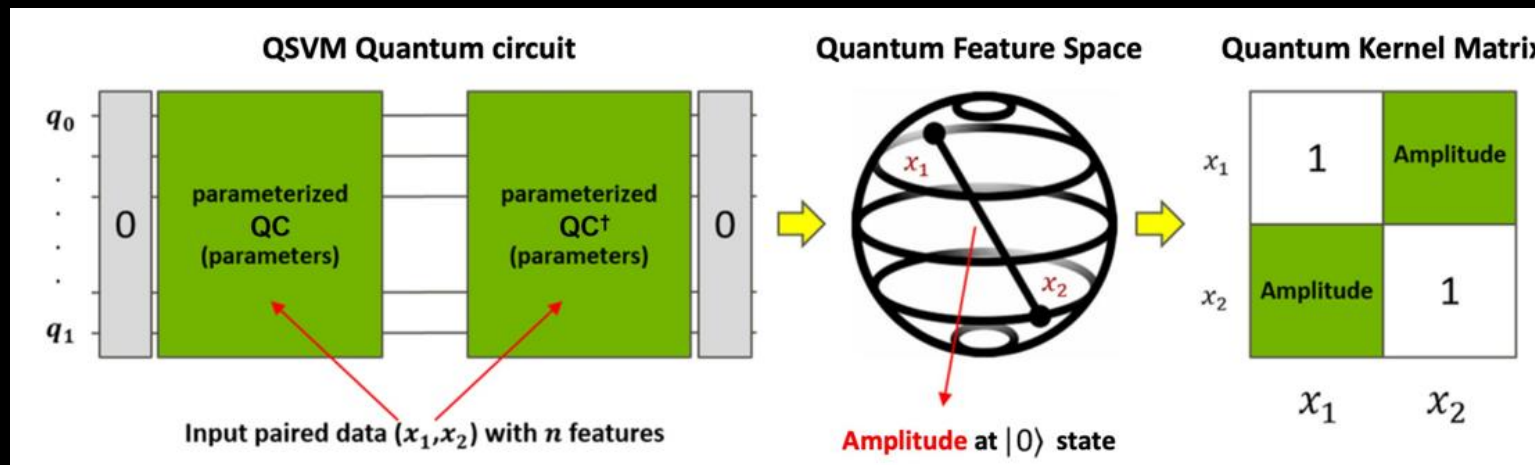


cuTN-QSVM 1.0

cuTensorNet-accelerated Quantum-enhanced Support Vector Machine

Issue 1 : Data Size

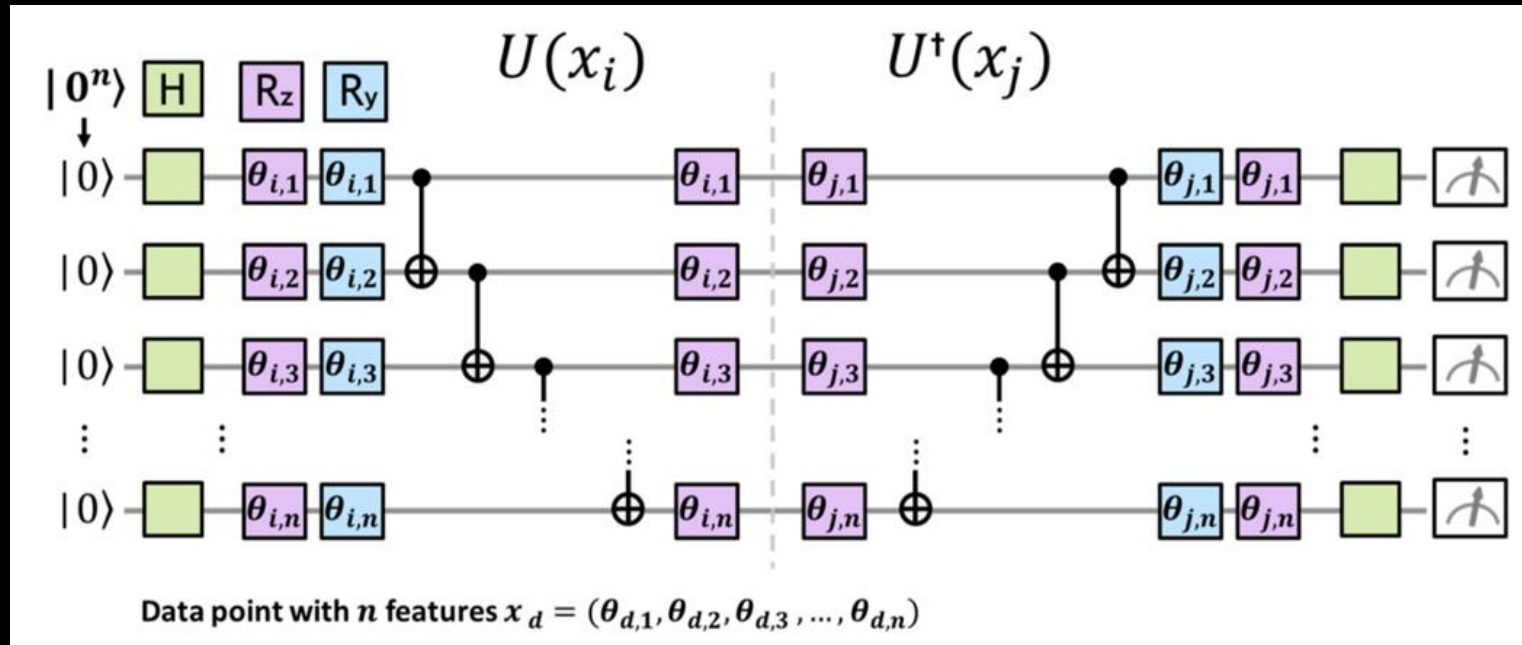
Quantum Support Vector Machine Workflow with Quantum Kernel



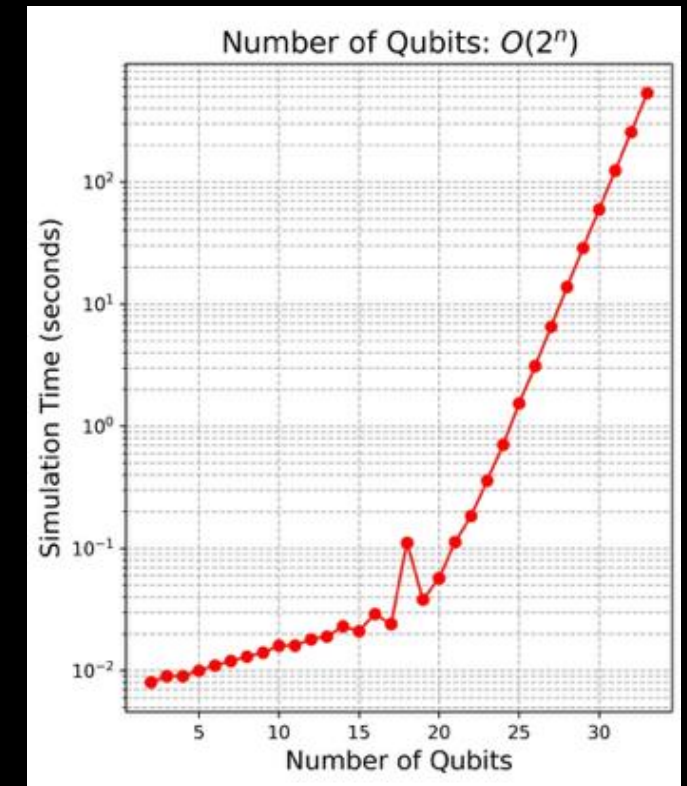
cuTN-QSVM 1.0

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Quantum Kernel Circuit



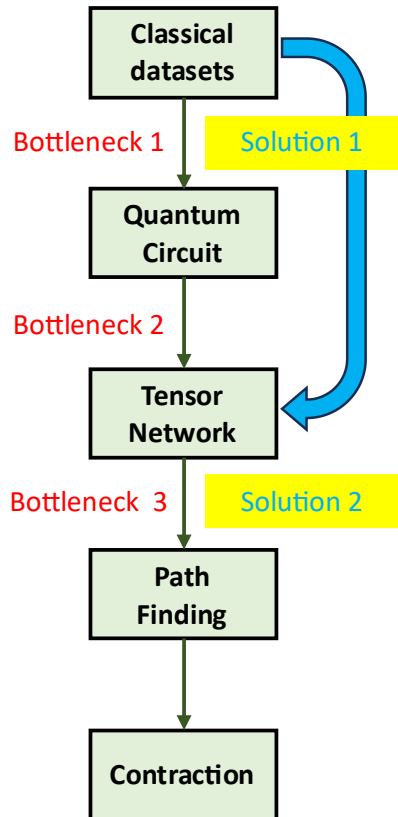
Issue 2 : Qubits



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cuTensorNet-accelerated Quantum-enhanced Support Vector Machine

Optimize Workflow



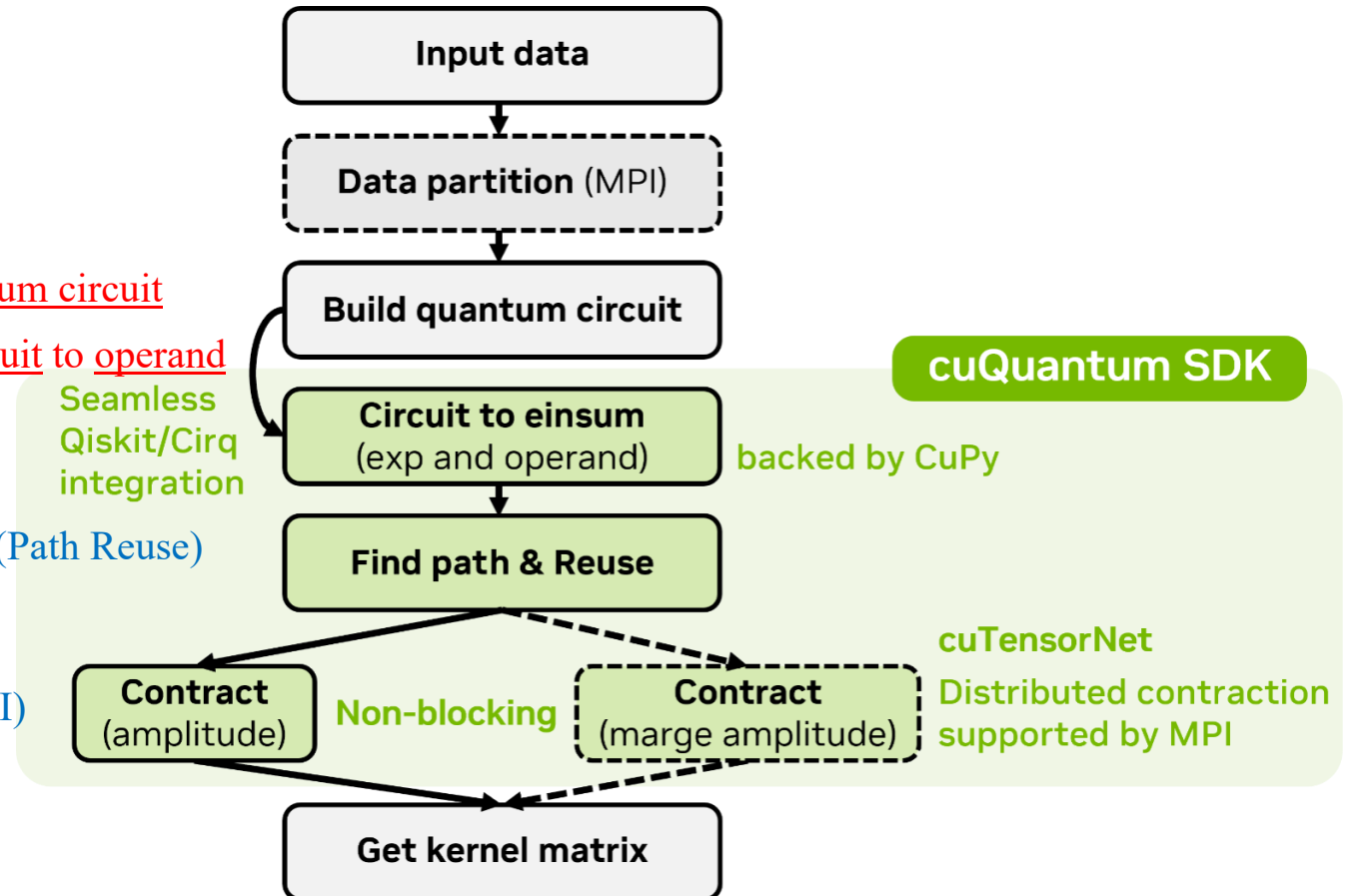
data to operand

Bottleneck1: data to quantum circuit

Bottleneck2: quantum circuit to operand

Bottleneck3: path finding (Path Reuse)

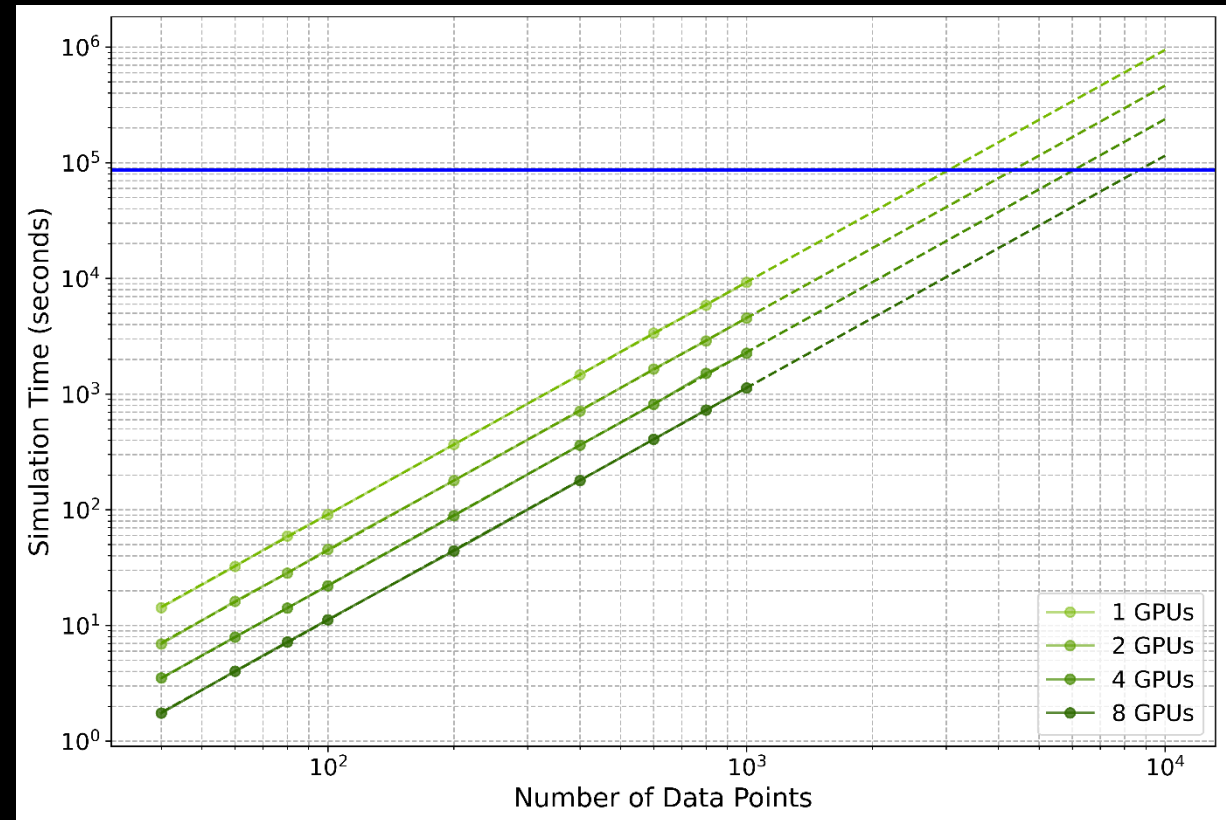
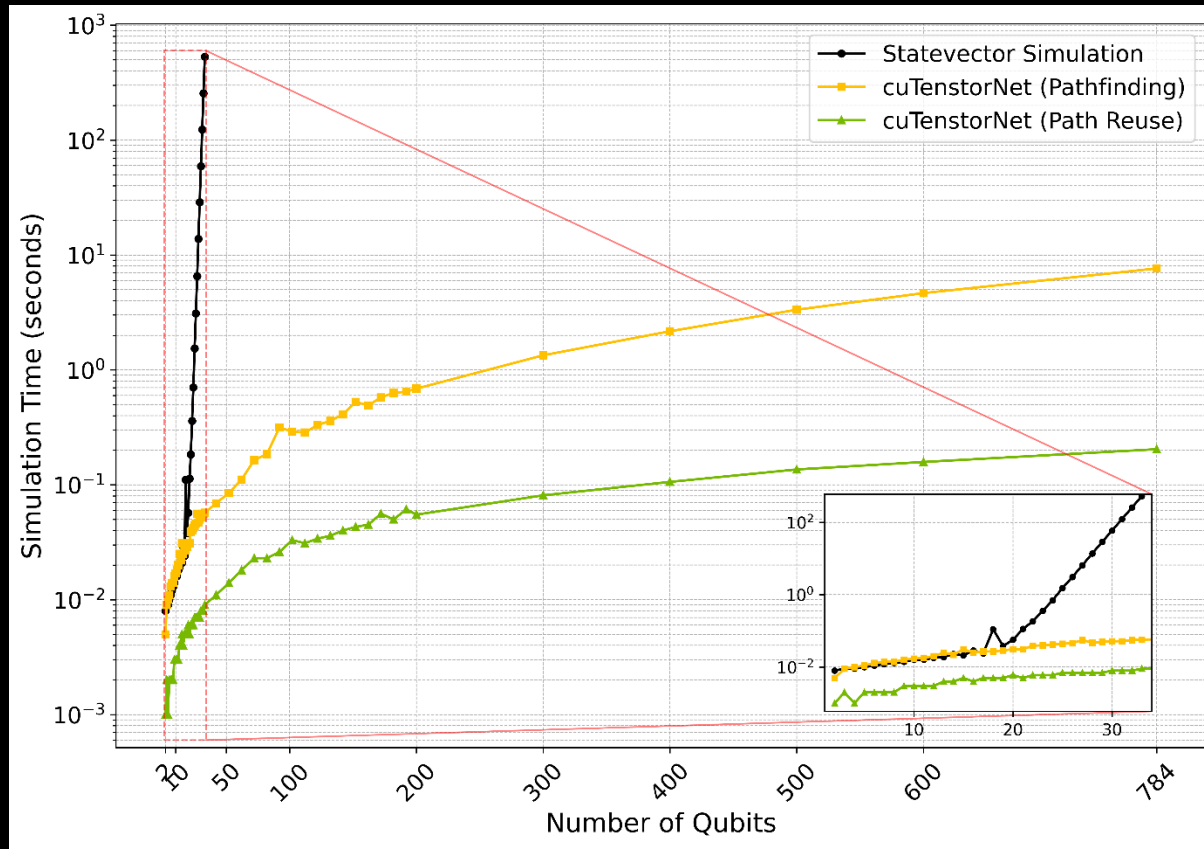
Bottleneck4: contract (MPI)



cuTN-QSVM 1.0

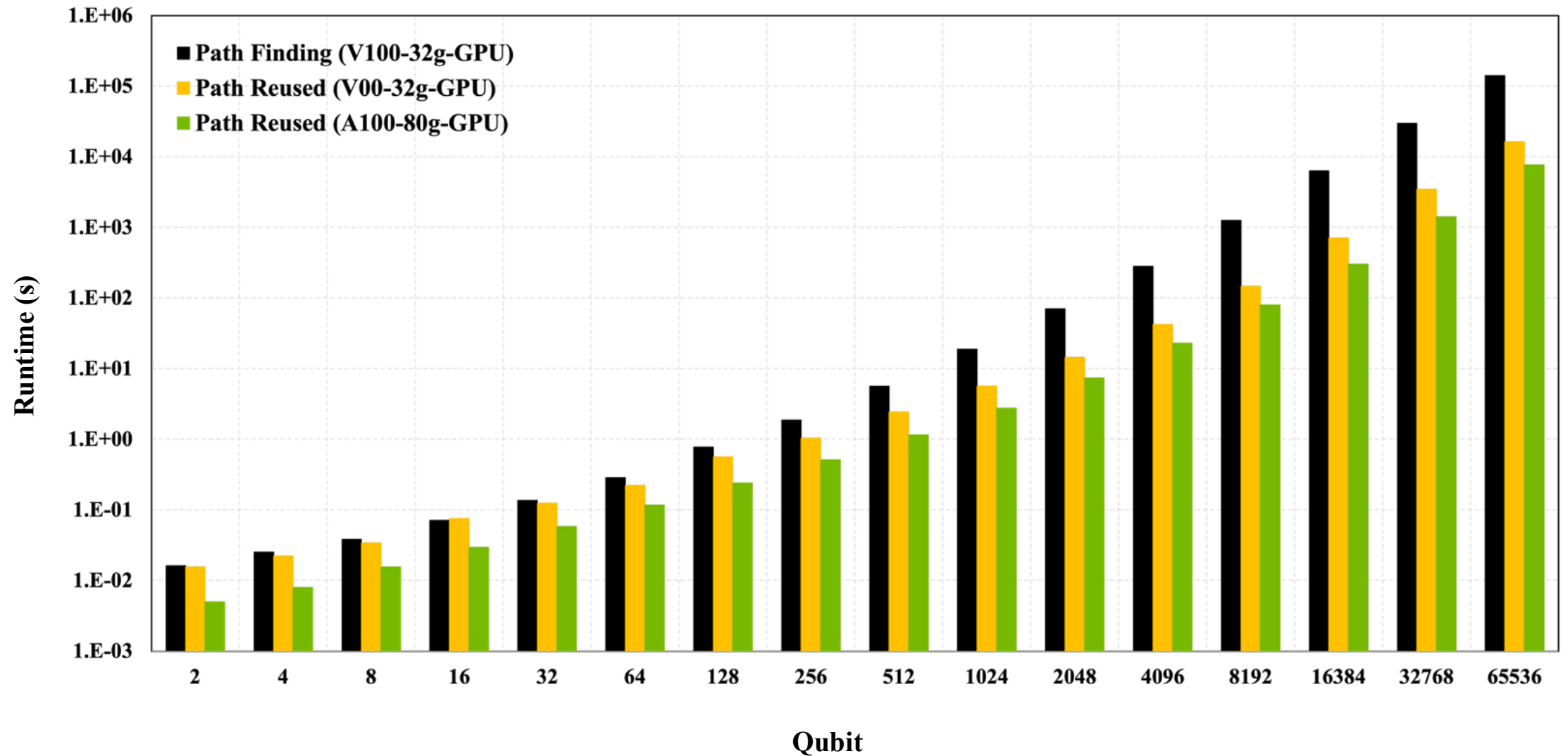
cuTensorNet-accelerated Quantum-enhanced Support Vector Machine

Single-GPU and Multi-GPUs Benchmark



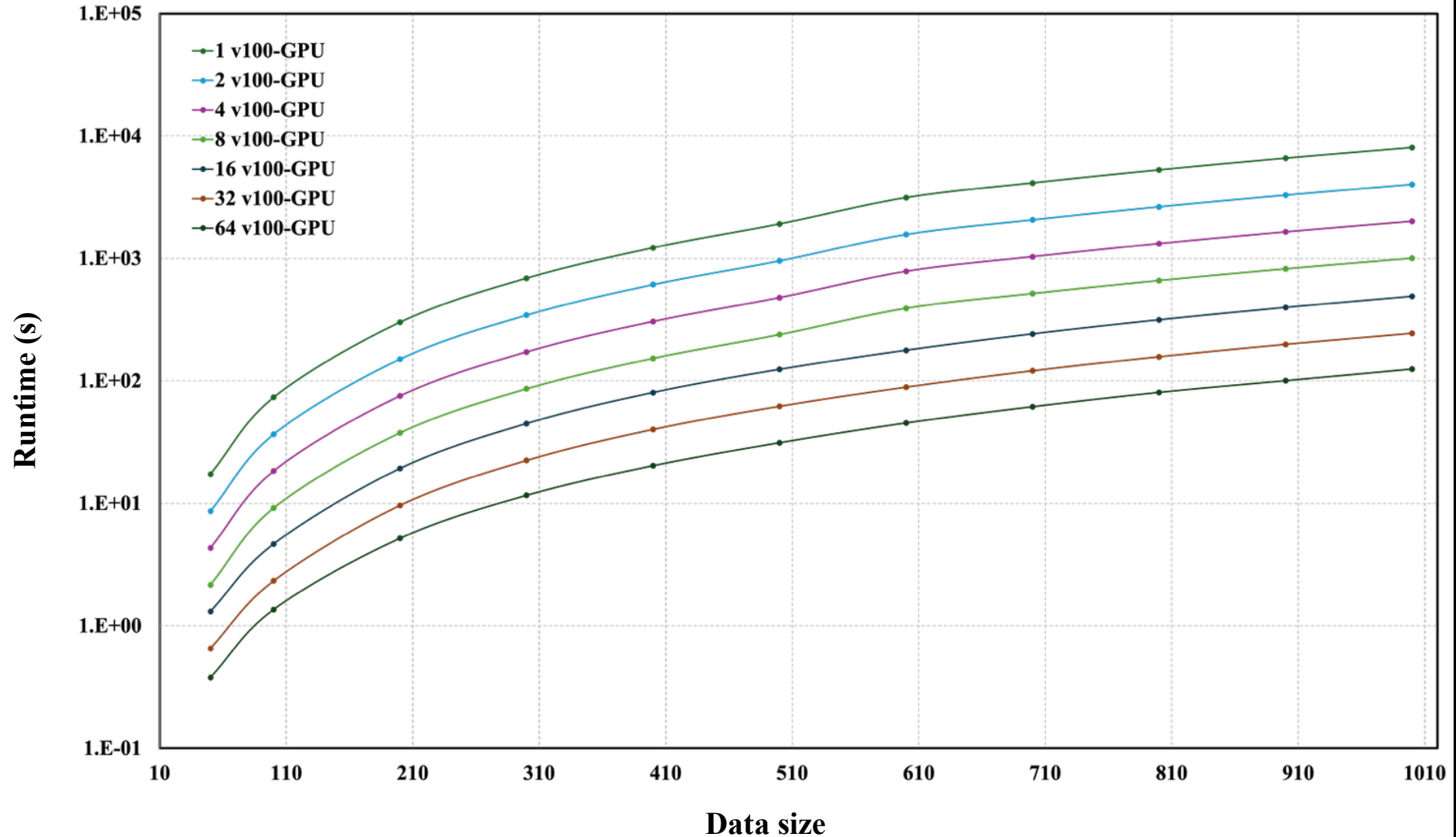
cuTN-QSVM 1.0

cuTensorNet-accelerated Quantum-enhanced Support Vector Machine



cuTN-QSVM 1.0

cuTensorNet-accelerated Quantum-enhanced Support Vector Machine

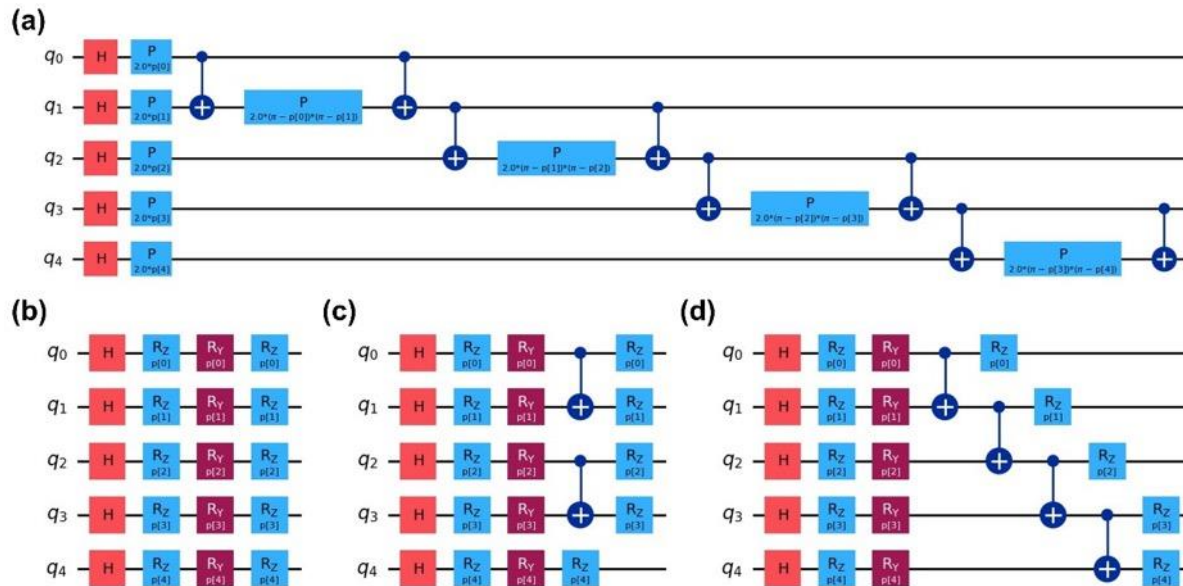


cuTN-QSVM 1.0

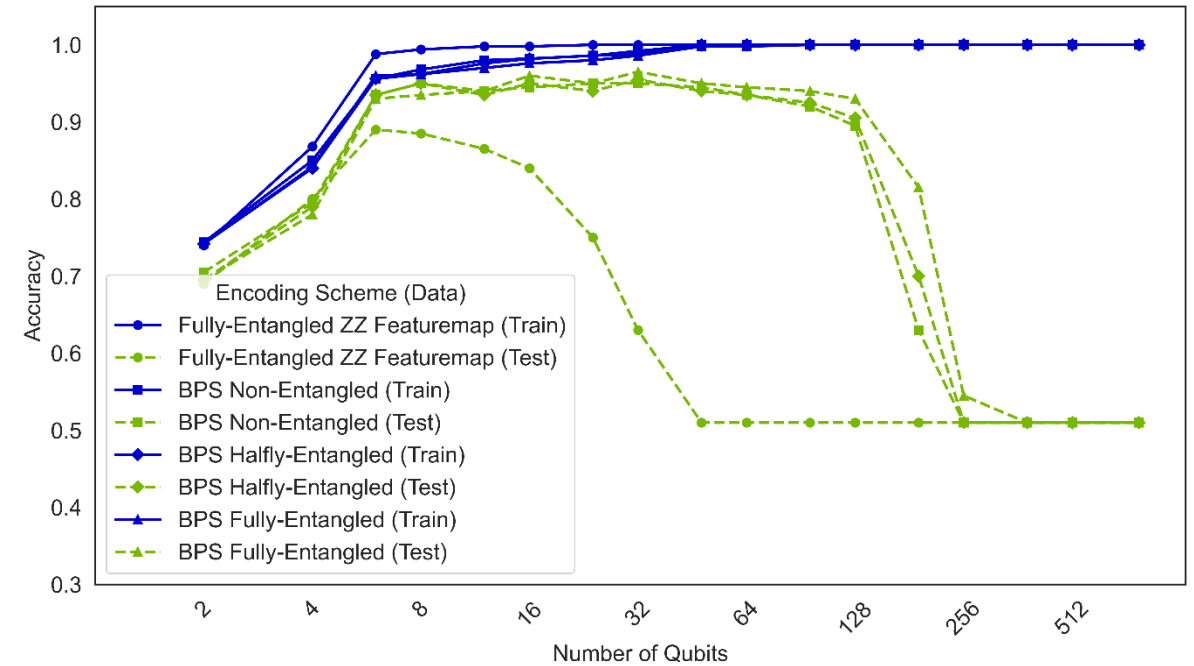
cuTensorNet-accelerated Quantum-enhanced Support Vector Machine

Performance of Quantum Circuit Architectures and Entanglement Levels

Different Quantum Circuit Architectures and Entanglement Levels

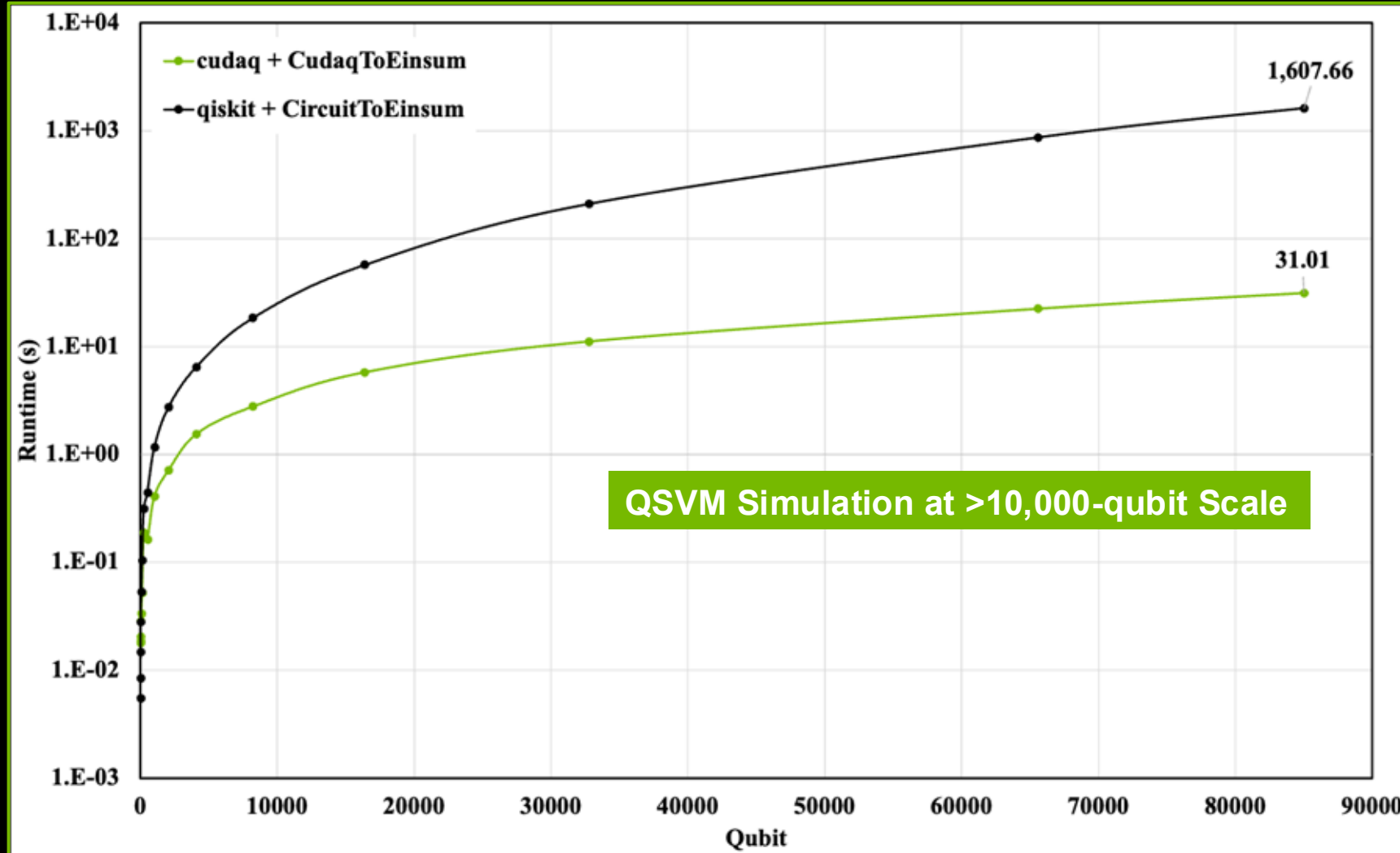


Performance



cuTN-QSVM 2.0

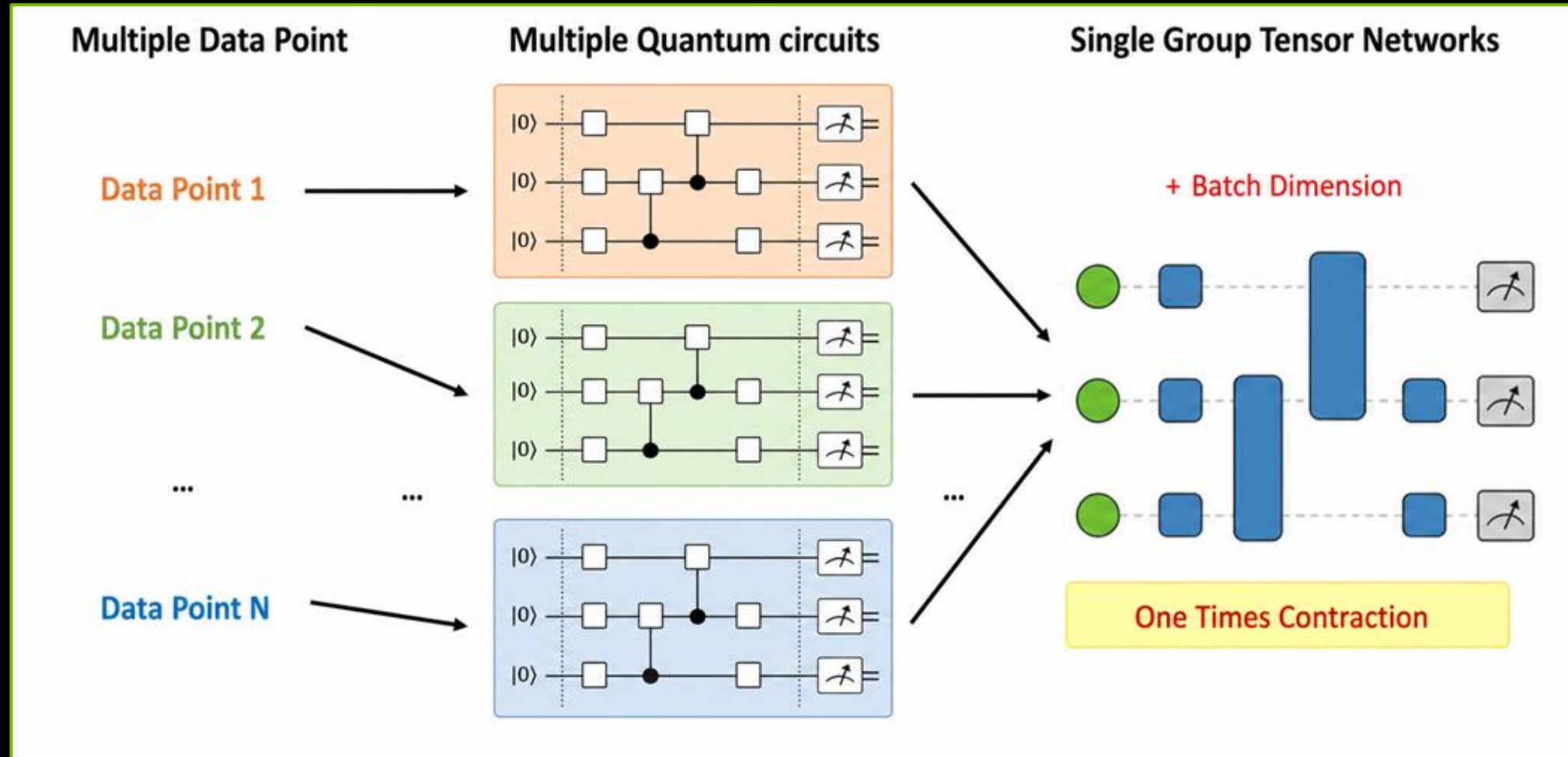
CudaqToEinsum: Efficient Tensor-Network Conversion from CUDA-Q



**x50
Speed up**

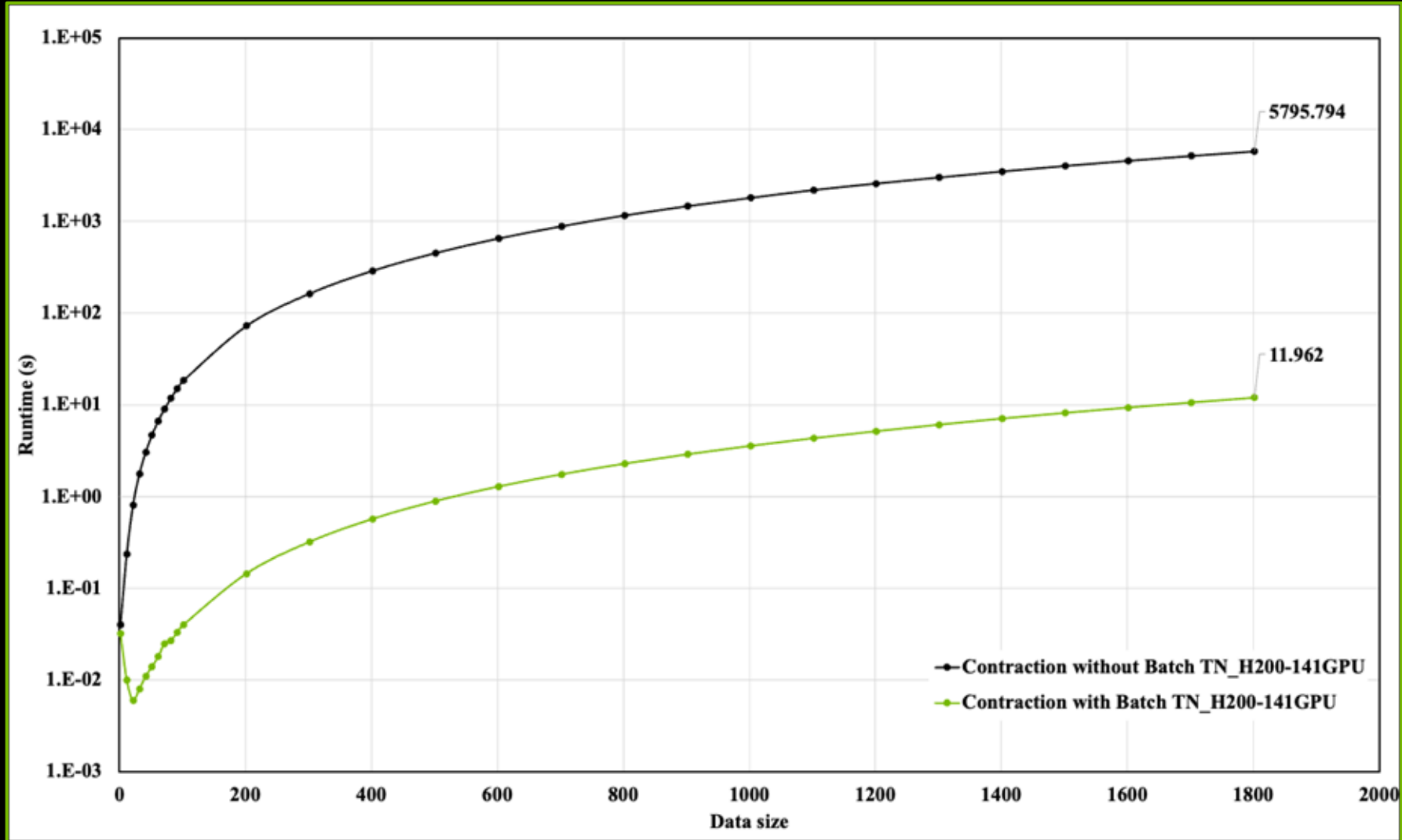
cuTN-QSVM 2.0

Efficient Batch Tensor-Network Simulation



cuTN-QSVM 2.0

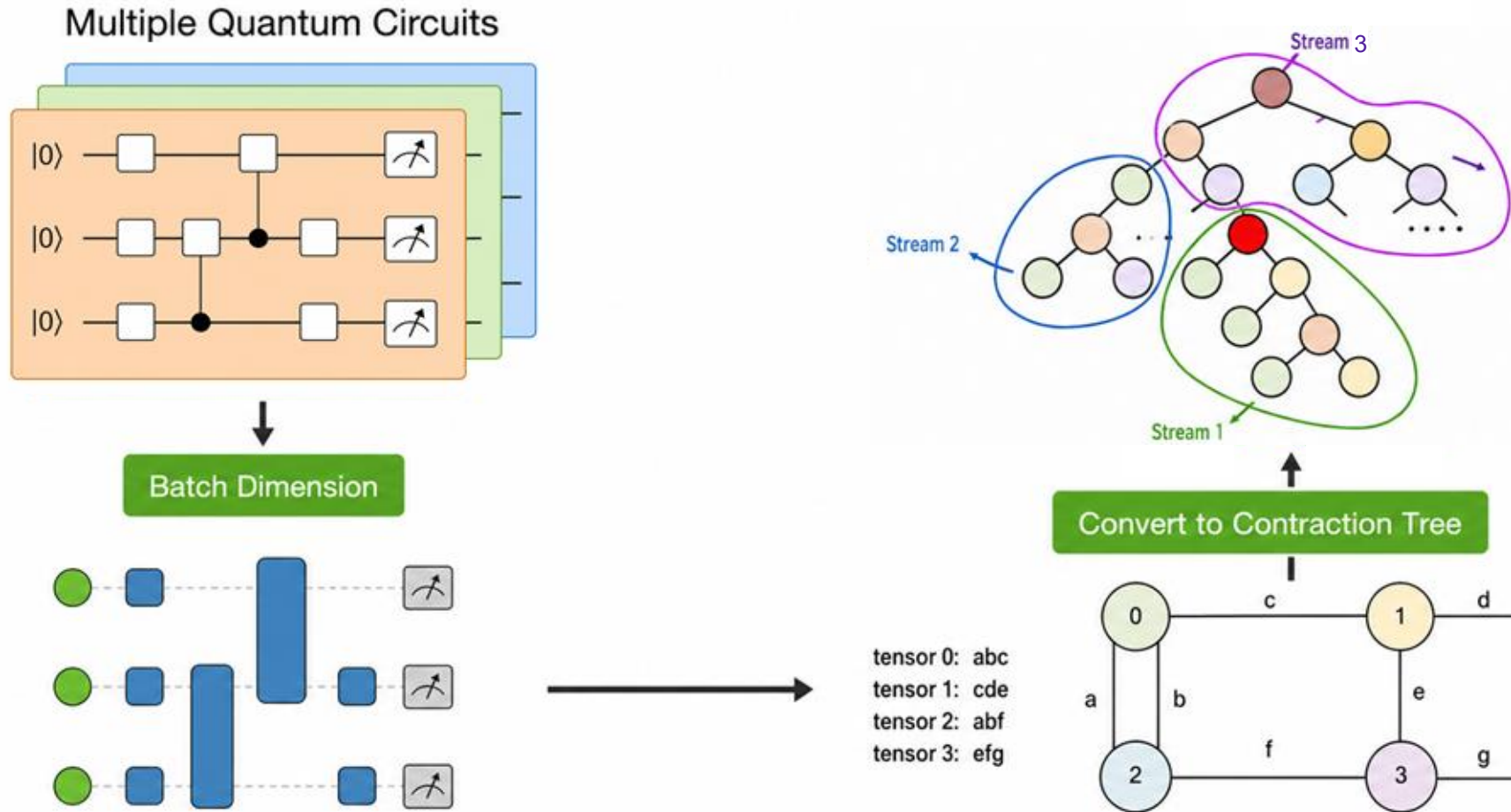
Efficient Batch Tensor-Network Simulation



x526
Speed up

cuTN-QSVM 2.0

Multi-stream Tensor-Network Simulation Optimized by cuTENSOR



Thanks

Tai-Yue Li

